**SCHOOL OF SCIENCE AND TECHNOLOGY**

DEPARTMENT OF MATHEMATICAL SCIENCES

BACHELOR OF SCIENCE IN MATHEMATICAL SCIENCES EDUCATION

ASSIGNMENT

SUBMITTED TO: DR. A. TAYLOR

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Question 1

A weighted, undirected graph G=(V,E) representing major towns across Malawi (20 nodes) with approximate driving distances in kilometers as edge weights.

Vertices (V,20 nodes)

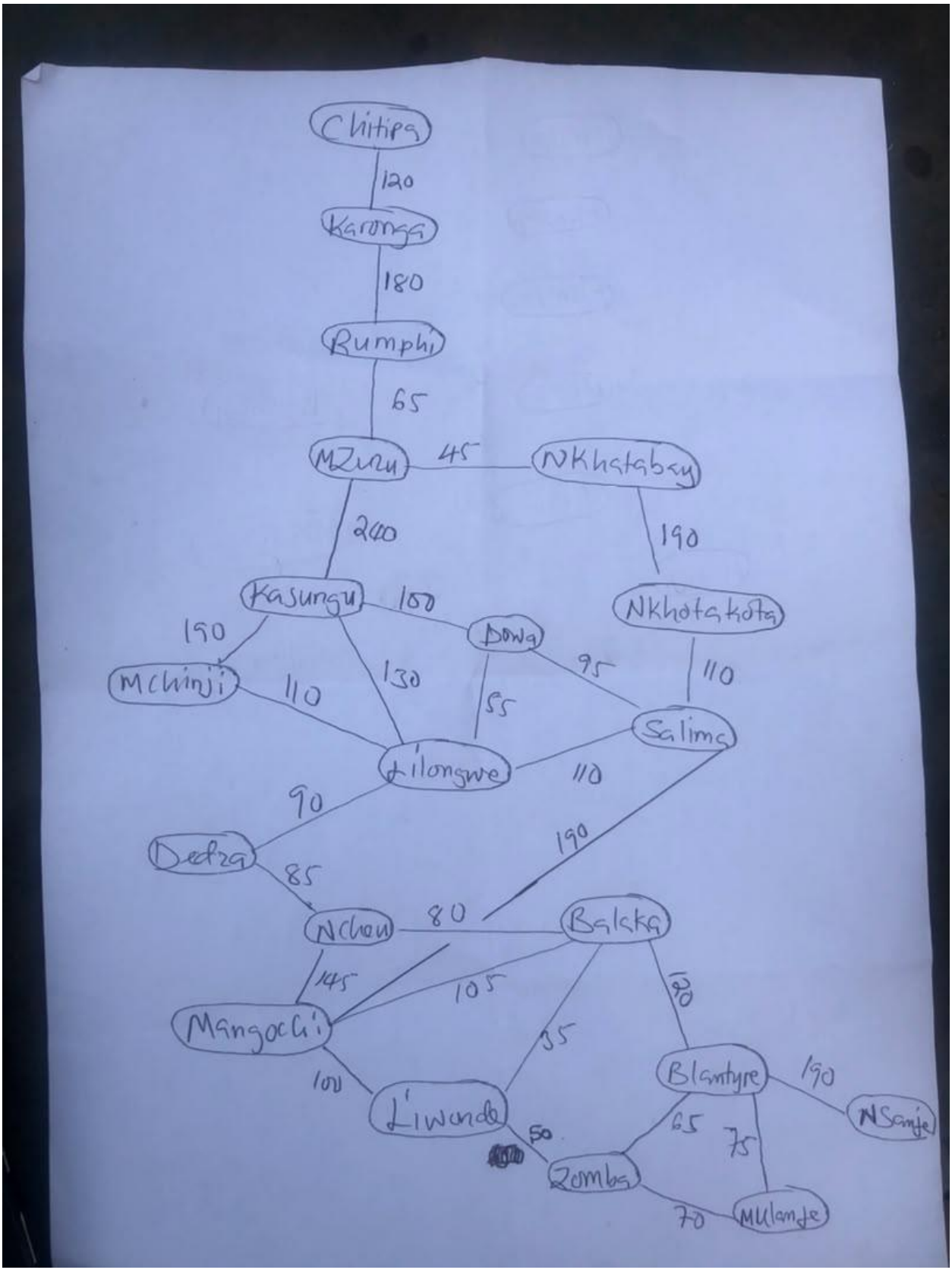
| Nodes in the graph | | | | |
|---|---|---|---|---|
| 1. Chitipa | 2. Karonga | 3. Rumphi | 4. Mzuzu | 5. Nkhatabay |
| 6. Kasungu | 7. Mchinji | 8. Lilongwe | 9. Dowa | 10. Nkhotakota |
| 11. Salima | 12. Dedza | 13. Ntcheu | 14. Mangochi | 15. Balaka |
| 16. Zomba | 17. Liwonde | 18. Blantyre | 19. Mulanje | 20. Nsanje |

Weighted edge(E)

| No. | Road connection | Weight km |
|---|---|---|
| 1. | Chitipa – Karonga | 120 |
| 2. | Karonga –Rumphi | 180 |
| 3. | Rumphi-Mzuzu | 65 |
| 4. | Mzuzu-Nkhatabay | 45 |
| 5. | Mzuzu-Kasungu | 240 |
| 6. | Nkhatabay-Nkhotakota | 190 |
| 7. | Kasungu-Dowa | 100 |
| 8. | Kasungu-Lilongwe | 130 |
| 9. | Kasungu-Mchinji | 190 |
| 10. | Mchinji-Lilongwe | 110 |
| 11. | Dowa-Lilongwe | 55 |
| 12. | Dowa-Salima | 95 |
| 13. | Nkhotakota-Salima | 110 |
| 14. | Salima-Lilongwe | 110 |

| | | |
|---|---|---|
| 15. | Lilongwe-Dedza | 90 |
| 16. | Dedza-Ntcheu | 85 |
| 17. | Ntcheu-Balaka | 80 |
| 18. | Ntcheu-Mangochi | 145 |
| 19. | Salima-Mangochi | 190 |
| 20. | Mangochi-Balaka | 105 |
| 21. | Mangochi-Liwonde | 100 |
| 22. | Balaka-Liwonde | 35 |
| 23. | Liwonde-Zomba | 50 |
| 24. | Zomba-Blantyre | 65 |
| 25. | Blantyre-Mulanje | 75 |
| 26. | Blantyre-Nsanje | 190 |
| 27. | Zomba-Mulanje | 70 |
| 28. | Balaka-Blantyre | 120 |

Question 2


Chitipa
120
Karonga
180
Rumphi
65
Mzuzu
45
Nkhatabay
240
190
Kasungu
100
Nkhotakota
190
Dowa
95
110
Mchinji
110
130
55
Salima
Lilongwe
110
90
190
Dedza
85
Ncheu
80
Balaka
145
105
120
Mangochi
55
Blantyre
190
100
Liwonde
50
65
75
Nsanje
Zomba
70
Mulanje

Graph discription

The graph represents simplified road network connecting 20 selected town in Malawi. It is modelled as undirected, weighted graph, represented as G= (V,E), where (V) represents the set of vertices(towns) and (E) represents the set of edges(road connections).

Each town is represented by vertex, drawn as circle, while each road connection is presented by an edges joining two vertices. The graph contains 20 vertices and 28 edges. Each edge is assigned a numerical weight representing the illustrative distance in kilometers between two connected towns.

The graph is undirected because movement between connected towns is considered possible in both directions. It is weighted because every road connection has distance value. Furthermore, the graph is connected because there is the path between every pair of towns, either directly or through other connected towns. It also contain circles because some towns can be reached through more than one possible root.

Question 3

The starting town is Lilongwe (8) and the destination is Blantyre (18).

Initialization

Dist[8]=0 all other Dist=∞

Visited={}

1. CURRENT = 8 (Lilongwe) Dist=0

   NewLen = Dist[8]+W(8,12)=0+90=90

   Dist[12]=90

2. CURRENT = 12 (Dedza) Dist=90

   NewLen = Dist+W(12,13)=90+85=175

   Dist[12]=175

   Next min is CURRENT = 13 (175)

3. CURRENT = 13 (Ntcheu) Dist=175

   For 15

   NewLen =175+W(13,15)=175+80=255

   Dist[15]=255

   For NE{14}

   NewLen =175+W(13,14)=175+90=265

   Dist[14]=265

   Min unvisited is 15 with 255 < 265 → CURRENT = 15 (255)

4. CURRENT = 15 (Balaka) Dist[15]=255 For NE{18 }

   NewLen =255+W(15,18)=255+120=375

   Dist[18]=375

   For NE{14}

   NewLen =255+60=315 > 265

   Dist[14]=265

   For NE{17}

   NewLen =255+65=320

   Dist[17]=320

Visited = {8,12,13,15}

We now have Dist[18]=375, Dist[14]=265, Dist[17]=320. Smallest is 14.

Path via 14→15→18 path is longer

Path via 15→17→16→18: 320+60+70 =450 >375

So min unvisited that leads to destination is still 18.

CURRENT = 18 (Blantyre) Dist[18]=375

5. TERMINATION

Destination vertex 18 is now CURRENT. STOP.

Lilongwe (8) → Dedza(12) → Ntcheu(13) → Balaka(15) → Blantyre(18)

QUESTION 5: TESTING THE ALGORITHM

Testing is the process of checking whether a program produces the expected results when different inputs are supplied. The program is tested using different starting and destination towns to check whether Dijkstra's Algorithm calculates the correct shortest distances.

A. Test cases

The following test cases are calculated from the 20-town road network.

| Test | Input: Start | Input: Destination | Expected distance |
|---|---|---|---|
| 1 | Lilongwe (8) | Blantyre (18) | 375 km |
| 2 | Lilongwe (8) | Dedza (12) | 90 km |
| 3 | Lilongwe (8) | Nsanje (20) | 565 km |
| 4 | Mzuzu (4) | Blantyre (18) | 745 km |
| 5 | Lilongwe (8) | Lilongwe (8) | 0 km |
| 6 | Chitipa (1) | Blantyre (18) | 1,110 km |

B. Sample input and expected output

Test 1: Lilongwe to Blantyre

Input:

8

18

Expected output:

Starting town: Lilongwe
Destination: Blantyre
Shortest distance: 375 km
Shortest path: Lilongwe -> Dedza -> Ntcheu -> Balaka -> Blantyre

Test 2: Lilongwe to Dedza

Input:

8
12

Expected output:

Starting town: Lilongwe
Destination: Dedza
Shortest distance: 90 km
Shortest path: Lilongwe -> Dedza

Test 3: Lilongwe to Nsanje

Input:

8
20

Expected output:

Starting town: Lilongwe
Destination: Nsanje
Shortest distance: 565 km
Shortest path: Lilongwe -> Dedza -> Ntcheu -> Balaka -> Blantyre -> Nsanje

C. Error testing

The program should also be tested with invalid inputs to check whether it handles errors correctly.

| Error test | Input | Expected program response |
|---|---|---|
| Invalid starting town | 25 | Invalid starting town number |
| Invalid destination | 30 | Invalid destination town number |
| Non-numeric input | ABC | Invalid input. Enter a number |
| Exit option | 0 | Thank you for using the program |

Testing was conducted to determine whether the implemented Dijkstra's Shortest Path Algorithm produces correct results when finding the shortest distances between different towns in Malawi. The program was tested using different starting and destination vertices from the given road network.

The main aim of testing was to check whether the algorithm correctly calculates the shortest distances and identifies the appropriate routes between towns.

C. Dry Run of Dijkstra's Shortest Path Algorithm

A dry run was conducted to demonstrate how Dijkstra's algorithm finds the shortest route from Lilongwe (Vertex 8) to Blantyre (Vertex 18).

| Step | Current town | Distance (km) | Calculation / Update | Updated town |
|---|---|---|---|---|
| 1 | Lilongwe (8) | 0 | Starting distance = 0 | Dowa = 55, Dedza = 90, Mchinji = 110, Salima = 110, Kasungu = 130 |
| 2 | Dowa (9) | 55 | 55 + 100 = 155; 55 + 95 = 150 | No update to Kasungu or Salima |
| 3 | Dedza (12) | 90 | 90 + 85 = 175 | Ntcheu = 175 |
| 4 | Mchinji (7) | 110 | No shorter distance found | No update |

| | | | | |
|---|---|---|---|---|
| 5 | Salima (11) | 110 | 110 + 110 = 220; 110 + 190 = 300 | Nkhotakota = 220, Mangochi = 300 |
| 6 | Kasungu (6) | 130 | 130 + 240 = 370 | Mzuzu = 370 |
| 7 | Ntcheu (13) | 175 | 175 + 80 = 255; 175 + 145 = 320 | Balaka = 255; Mangochi remains 300 |
| 8 | Nkhotakota (10) | 220 | 220 + 190 = 410 | Nkhatabay = 410 |
| 9 | Balaka (15) | 255 | 255 + 120 = 375; 255 + 35 = 290 | Blantyre = 375, Liwonde = 290 |
| 10 | Liwonde (17) | 290 | 290 + 50 = 340 | Zomba = 340 |
| 11 | Mangochi (14) | 300 | No shorter distance found | No update |
| 12 | Zomba (16) | 340 | 340 + 65 = 405; 340 + 70 = 410 | No shorter update to Blantyre or Mulanje |
| 13 | Mzuzu (4) | 370 | 370 + 65 = 435 | Rumphi = 435 |
| 14 | Blantyre (18) | 375 | Destination reached | Stop |

Final shortest route:

Lilongwe → Dedza → Ntcheu → Balaka → Blantyre.

Total distance: 90+85+80+120=375km

Question 6

Does the given graph have a Minimum Spanning Tree (MST)? If yes, what does it represent? Explain your answer.

Yes,

The graph has 20 vertices representing major towns in Malawi and 28 edges representing road connections, with each edge having a distance measured in kilometres.

The reasons why the graph has a Minimum Spanning Tree are explained below:

The graph is connected

The graph is connected because every town can be reached from any other town either directly or through other towns. For example, a person can travel from Chitipa to Blantyre through Karonga, Rumphi, Mzuzu and other connected towns. This means that no town is completely isolated. A connected graph allows an MST to include all vertices.

The graph is undirected

The graph is undirected because the road connections allow movement in both directions. For example, the road between Lilongwe and Dedza allows movement from Lilongwe to Dedza and from Dedza to Lilongwe. This makes the graph suitable for finding an MST using algorithms such as Kruskal's or Prim's algorithm.

The graph is weighted

The graph is weighted because each road connection has a numerical value representing the approximate distance in kilometres. For example, the connection between Balaka and Liwonde has a weight of 35 km, while the connection between Chitipa and Karonga has a weight of 120 km. These weights allow the algorithm to compare road connections and select edges that produce the minimum total distance.

The graph contains alternative routes

Some towns have more than one possible road connection. For example, Blantyre can be reached through Balaka or Zomba. These alternative connections provide different possible ways of connecting the towns. The MST algorithm compares the available edge weights and selects suitable connections without creating cycles.

The MST represents the minimum-total-distance combination of existing road connections needed to connect all 20 selected towns in Malawi. For example, it can help in identifying an efficient network of connections that links the towns while minimizing the sum of the selected road distances.

Question 7

I would use Kruskal's Algorithm to find the Minimum Spanning Tree (MST) of the given road network connecting the 20 towns in Malawi.

Kruskal's Algorithm is suitable because the given graph is an undirected, weighted and connected graph. The vertices represent the 20 towns, while the edges represent roads connecting the towns. The weights assigned to the edges represent the distances between the towns in kilometres.

Reasons for choosing Kruskal's Algorithm

1. It finds the Minimum Spanning Tree: Kruskal's Algorithm is specifically designed to find an MST by connecting all vertices using the minimum possible total edge weight.

2. It minimises the total distance: The algorithm selects roads with smaller distances first, helping to minimise the total distance of the selected road network.

3. It is suitable for weighted graphs: Our graph has 28 roads, each with a distance in kilometres. Kruskal's Algorithm uses these weights to determine which roads should be selected.

4. It connects all towns: The main objective is to ensure that all 20 towns are connected, either directly or indirectly, through the selected roads.

5. It avoids cycles: The algorithm checks that adding a road does not create a cycle. This ensures that the resulting network is a tree.

6. It selects edges systematically: It first arranges all roads in ascending order according to their distances and then considers the shortest roads first.

7. It produces the required number of edges: Since our graph contains 20 vertices, the MST must contain exactly 19 edges to connect all towns without cycles.

8. It is easy to understand and implement: The steps involve sorting the edges, selecting suitable roads and checking for cycles, making it relatively straightforward to implement.

9. It is appropriate for the given network: The graph has 28 edges connecting 20 towns. Kruskal's Algorithm can examine these edges and select the ones that provide the minimum total distance while maintaining connectivity.

10. It is more appropriate than Dijkstra's Algorithm for this task: Dijkstra's Algorithm finds the shortest path from a specified starting town to other towns. For example, it was used to find the shortest route from Lilongwe to Blantyre. Kruskal's Algorithm, on the other hand, finds the minimum total distance required to connect the entire network

Actual MST result for our graph

Using Kruskal's Algorithm, the following 19 edges are selected:

| Selected road | Distance (km) |
|---|---|
| Balaka – Liwonde | 35 |
| Mzuzu – Nkhatabay | 45 |
| Liwonde – Zomba | 50 |
| Dowa – Lilongwe | 55 |
| Rumphi – Mzuzu | 65 |
| Zomba – Blantyre | 65 |
| Zomba – Mulanje | 70 |
| Ntcheu – Balaka | 80 |
| Dedza – Ntcheu | 85 |
| Lilongwe – Dedza | 90 |
| Dowa – Salima | 95 |
| Kasungu – Dowa | 100 |

| | |
|---|---|
| Mangochi – Liwonde | 100 |
| Mchinji – Lilongwe | 110 |
| Nkhotakota – Salima | 110 |
| Chitipa – Karonga | 120 |
| Karonga – Rumphi | 180 |
| Nkhatabay – Nkhotakota | 190 |
| Blantyre – Nsanje | 190 |
| Total | 1,835 km |

Calculated Minimum Spanning Tree

1,835 km

Vertices connected

20 towns

Selected edges

19 roads

Cycles in the MST

0